Kairos AI

Agentic reliability for manufacturing: catch the critical moment before downtime

- Theme: Manufacturing | Intelligent Operations & Industrial Efficiency.

- For plant reliability teams: onboard any device, learn from its documents and data, and turn telemetry into cited, actionable diagnoses.

- Built during the AI Builder Cup on Google Cloud: Gemini on Vertex AI, Vertex embeddings, Firestore, Cloud Storage, BigQuery, Secret Manager and Cloud Run.

The problem

Diagnosis is the bottleneck, not detection

- Alarms tell an engineer what changed, not what to review next; the knowledge is scattered across manuals, registers and event logs.

- One upstream fault becomes many tickets, and triage stays slow and tribal.

- Hypothesis: grounded evidence, an agentic engine and a cited engineer handoff cut triage effort - to be proven in an approved pilot, not assumed.

What it does

Onboard -> Learn -> Analyze -> Act, live or historic

- Devices: register any asset and upload datasheets and registers; retrieval runs over Vertex embeddings with a per-device Firestore corpus (RAG).

- Agentic chat with Ask / Plan / Agent modes: it clarifies, plans, then generates datasets, test cases, code and live-monitoring plans.

- Five-agent engine (Monitor -> Diagnose -> Decide -> Act -> Verify) scores uploaded datasets and live REST streams through one code path.

- Live monitoring: the agent reads the datasheet to pick parameters and thresholds, decodes event logs, and streams readings into the same engine.

Google Cloud architecture

Real services, service-account identity, no embedded keys

- Cloud Run: deployed app, scale-to-zero, runs as a dedicated service account via ADC.
- Gemini on Vertex AI: grounded reasoning, generation and structured output.
- Vertex embeddings + Firestore: per-device knowledge corpus and durable platform state (users, devices, sources, chunks).
- Cloud Storage + BigQuery: telemetry and datasets, plus a partitioned audit table with 30-day retention.

Innovation and honesty

Beyond a chatbot on a dashboard

- Datasheet-driven monitoring: the agent derives parameter IDs, thresholds and event-log decoding from the device's own documents.

- One engine, two modes: identical diagnosis for historic datasets and for a live warm-up then stream.

- Physically-consistent synthetic data with ground-truth scenarios scores diagnosis accuracy, downtime and cost avoided.

- Honest by design: offline fallbacks are labeled and never shown as Gemini; data is synthetic; humans approve actions.

Impact, feasibility and next step

Start with one approved workflow at one plant

- Impact: faster triage, consolidated tickets, and device knowledge made accessible to every engineer; scales device-by-device and site-by-site.

- Recorded live smoke test: 4/4 accepted Gemini diagnoses. This is not accuracy on a labeled production dataset.

- Pilot measures manual-vs-assisted triage time, engineer acceptance and appropriate escalation; no invented savings or market statistics.

- Submission: deployed link + public repository + this deck + a video strictly under three minutes.